

A.I. Matrix:

Authentic v. Synthetic Image Generation

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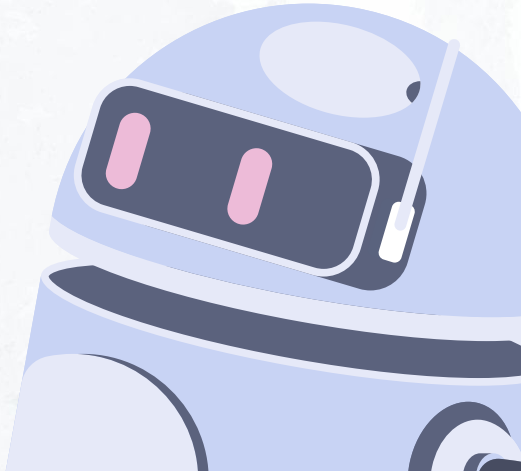


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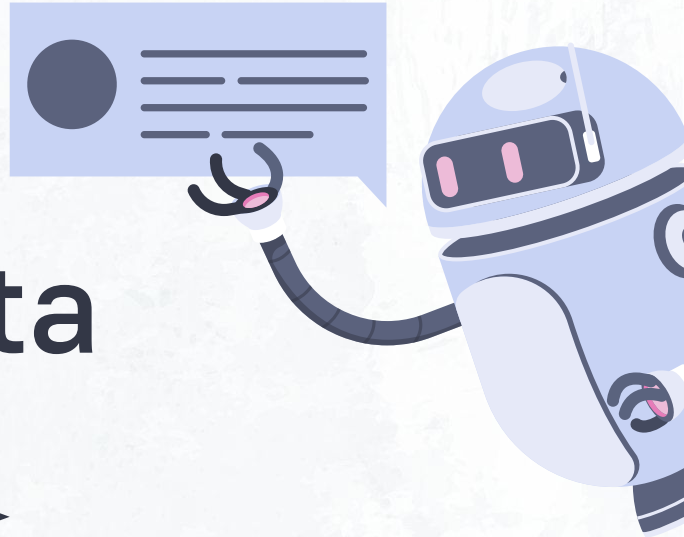
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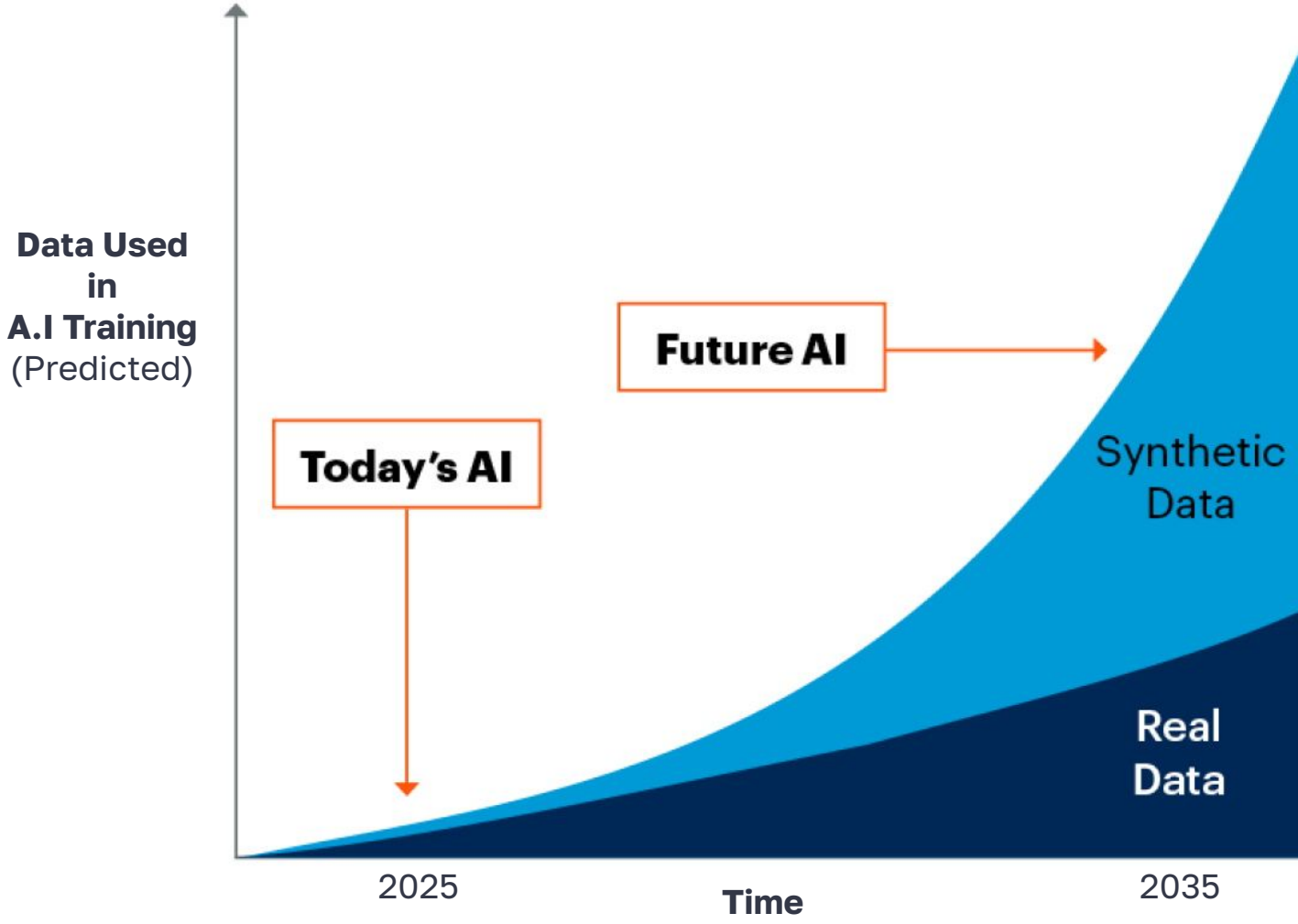
Introduction

AI & Synthetic Data



(AI)

THE FUTURE OF SYNTHETIC



Advantages & Disadvantages

(+) Unlimited Production

(-) Diversity/Bias

(+) Availability

(-) MAD





02

Frameworking & Datasets



03

Project Overview

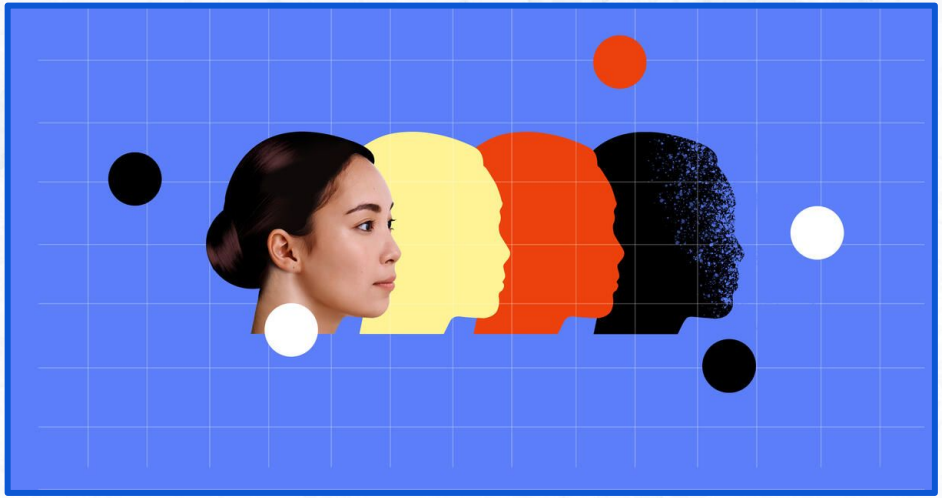
Datasets

(a) Synthetic Data →

What we looked for?

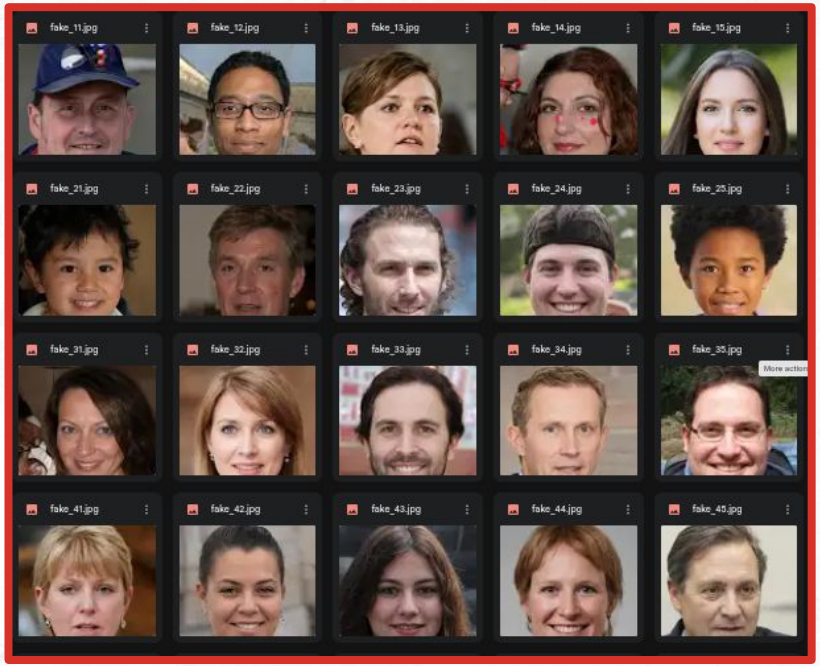
(b) Authentic Data →

How to compare?

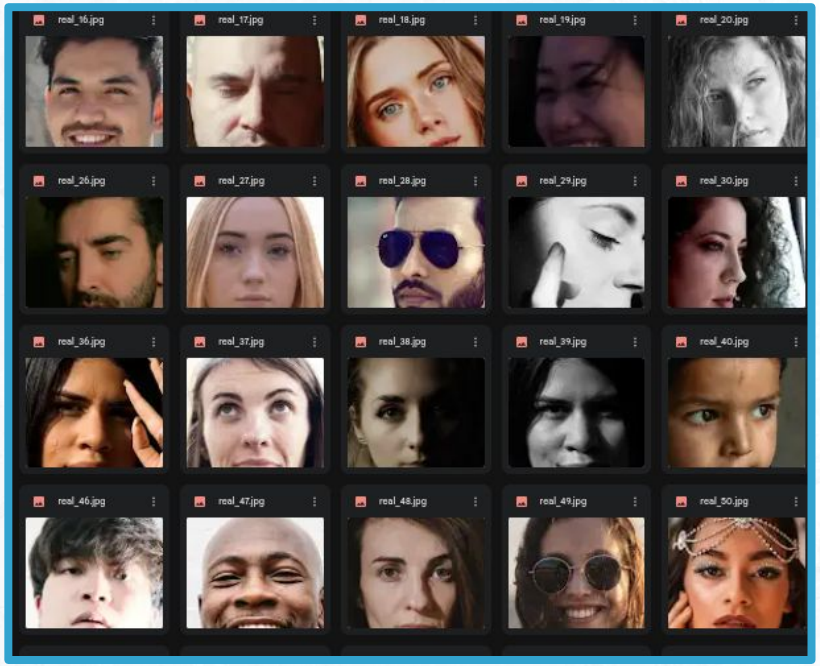


Generation

Fake

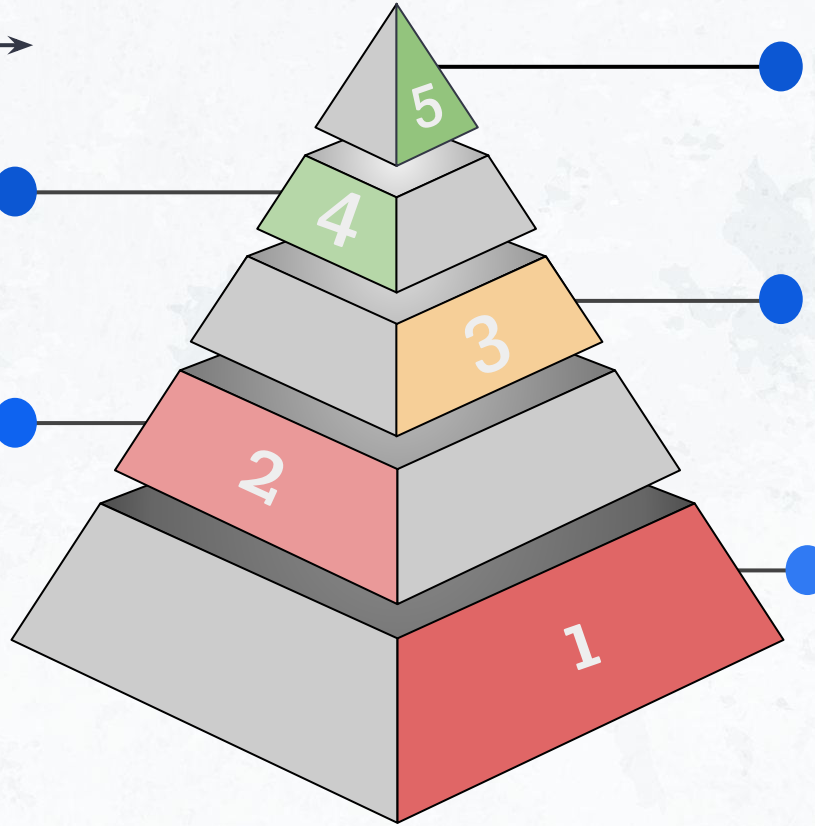


Real



Analysis Metric

(a) Likert Scale →



Excellent
Indistinguishable

Good
Few Errors

Average
Minimum requirements

Subpar
Barley accomplishes task

Horrible
Unrecognizable

Dataset Sample





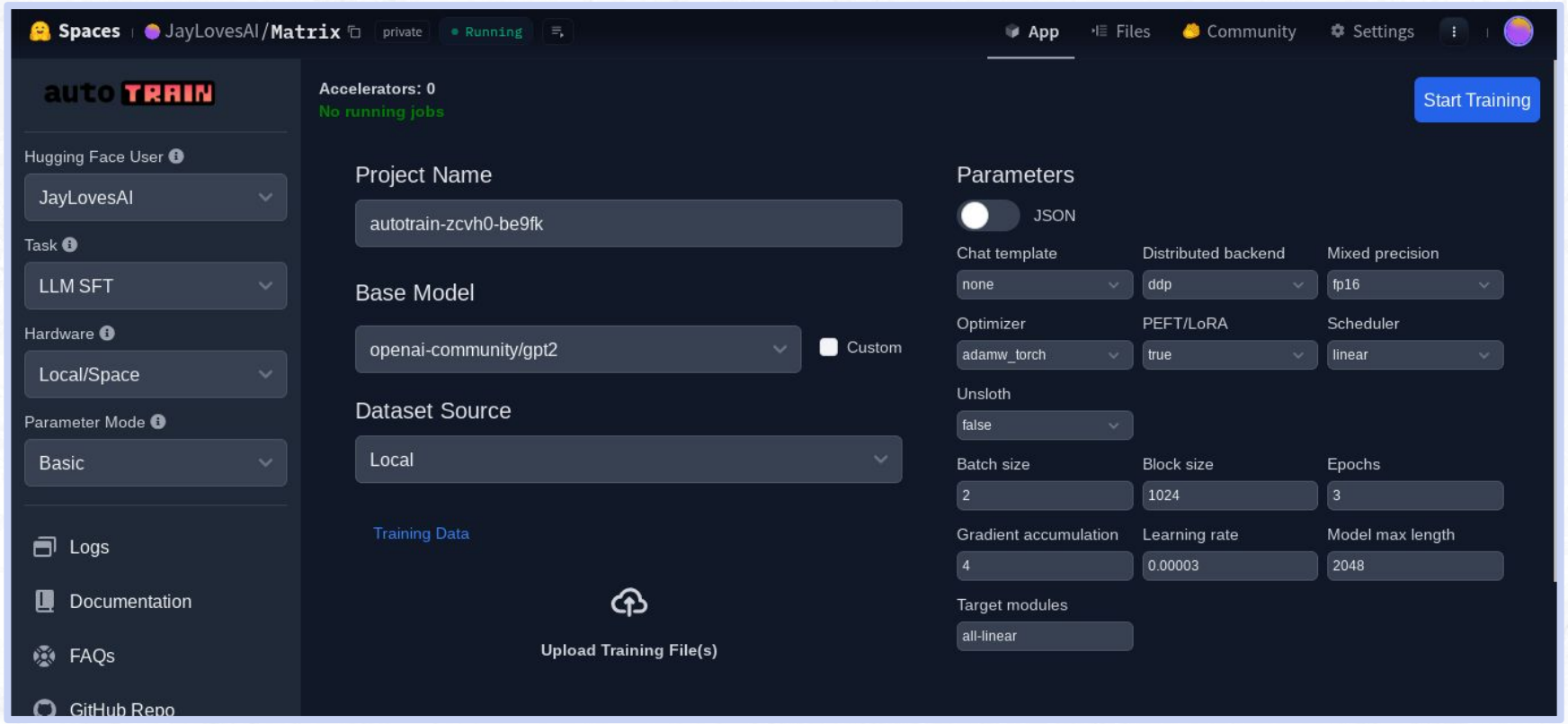
REAL

FAKE

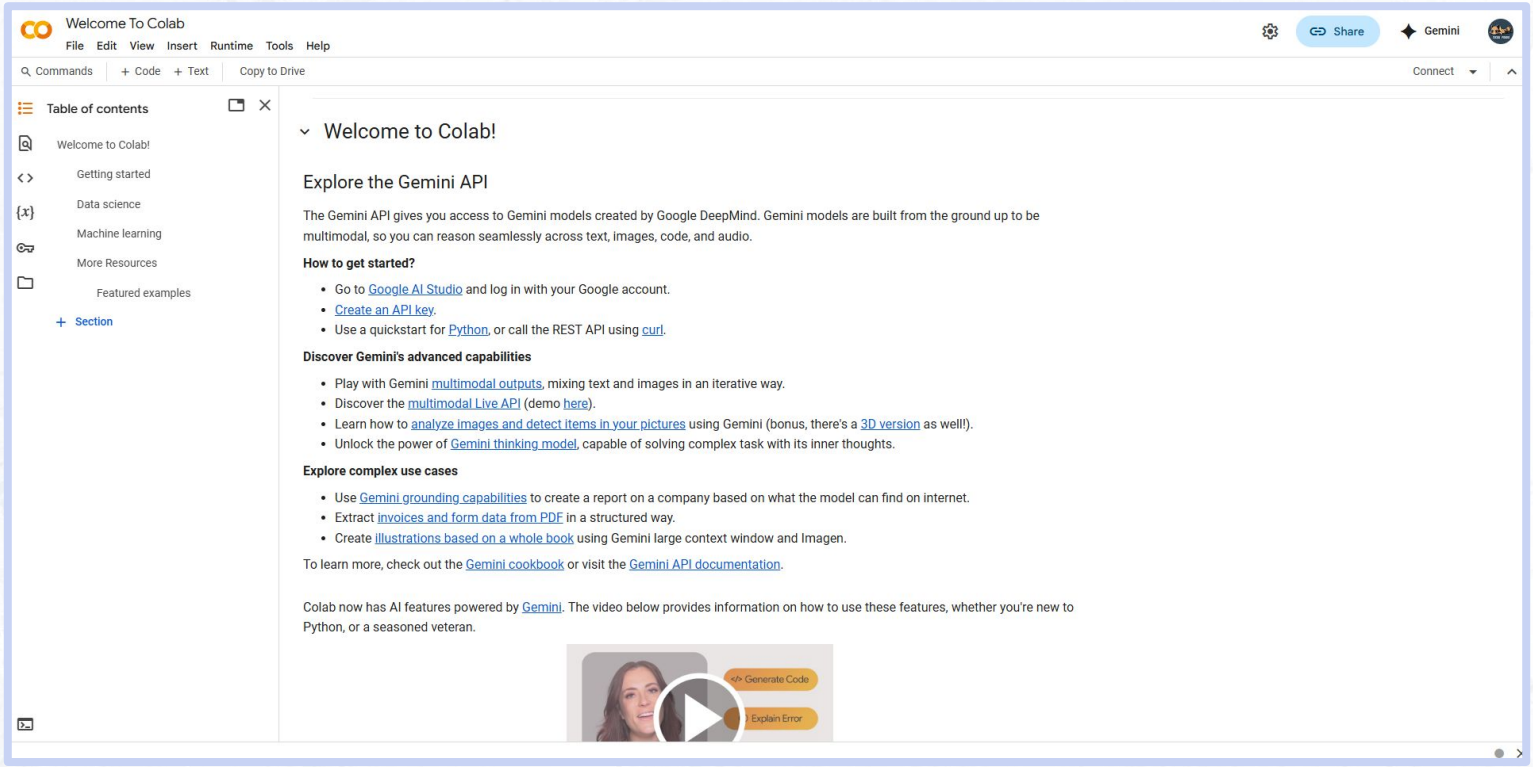




Generation Issues

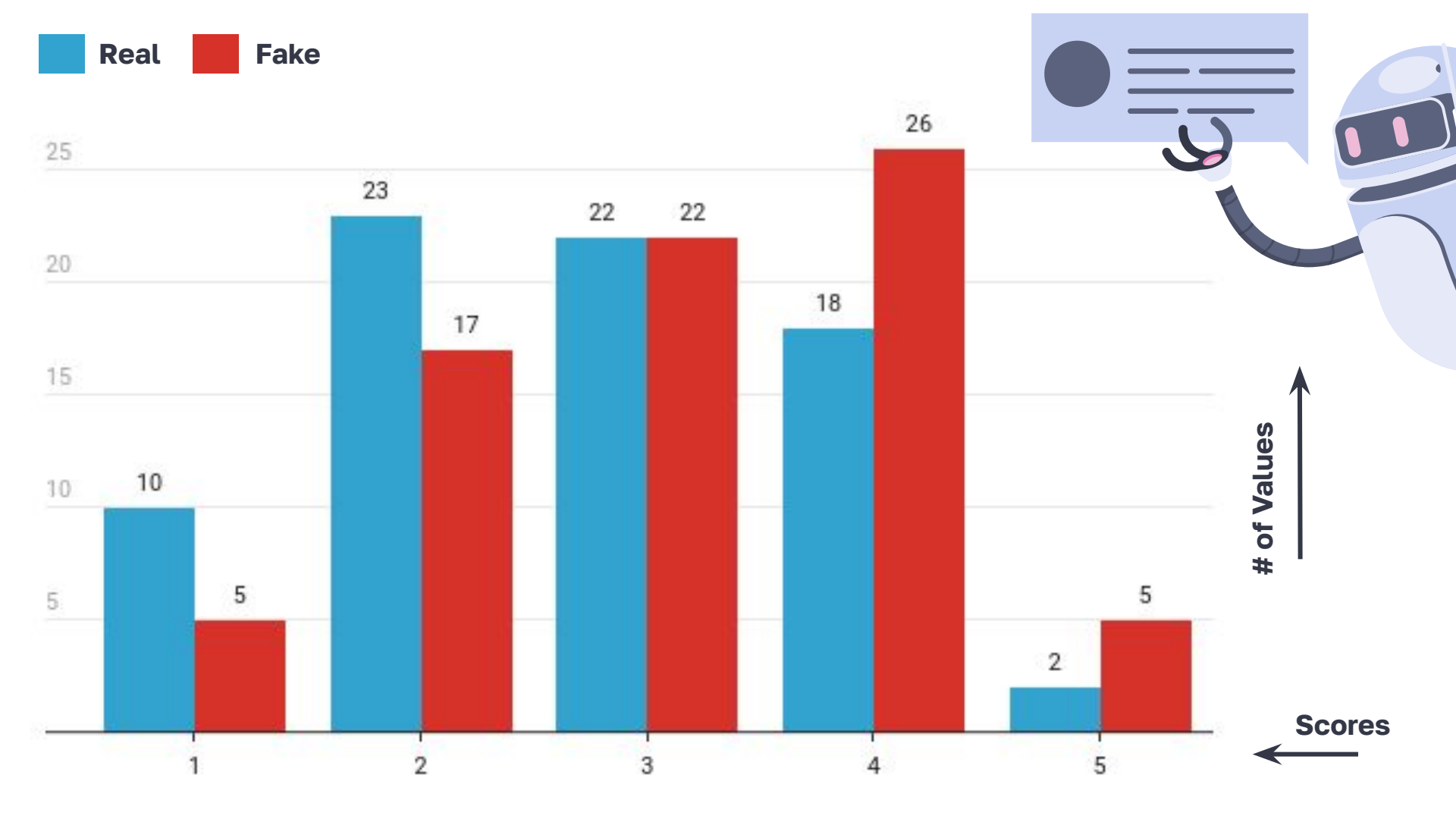


The Solution



04

Results


Real
Fake
25
20
15
10
5
10
5
23
17
22
22
18
26
2
5
1
2
3
4
5
of Values
Scores

REAL SCORES



2.72

(AI)

FAKE SCORES



3.12

(AI)

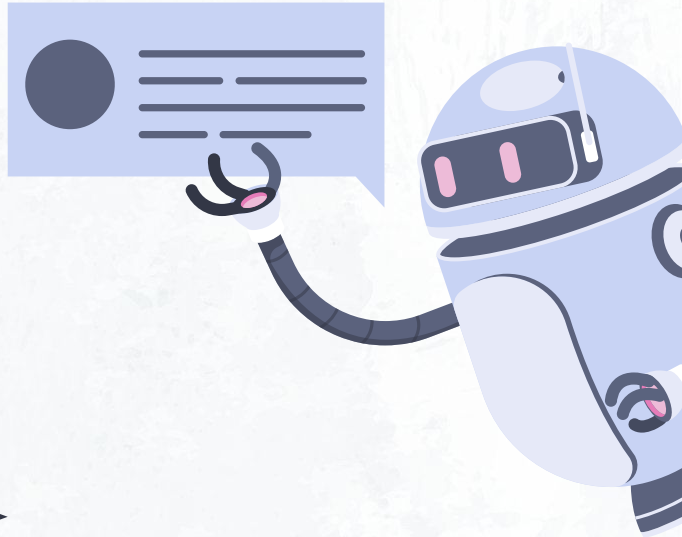
Significance?

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

0.0114

(AI)

Why Does This Matter?



(AI)

Thanks!

Any questions?

